Technical University of Munich
School of Computation, Information, and Technology
Department of Computer Science, SS 2026

Operations Research

Prof. Dr. Martin Bichler,
Halil Bayrak, Janik Bürgermeister, Teodora Dobos,
Jan Höhener, Abheek Ghosh

Sample solution to the demo tasks 6

Demo Exercises

Exercise D6. 1. *Branch-and-Bound*

Problem

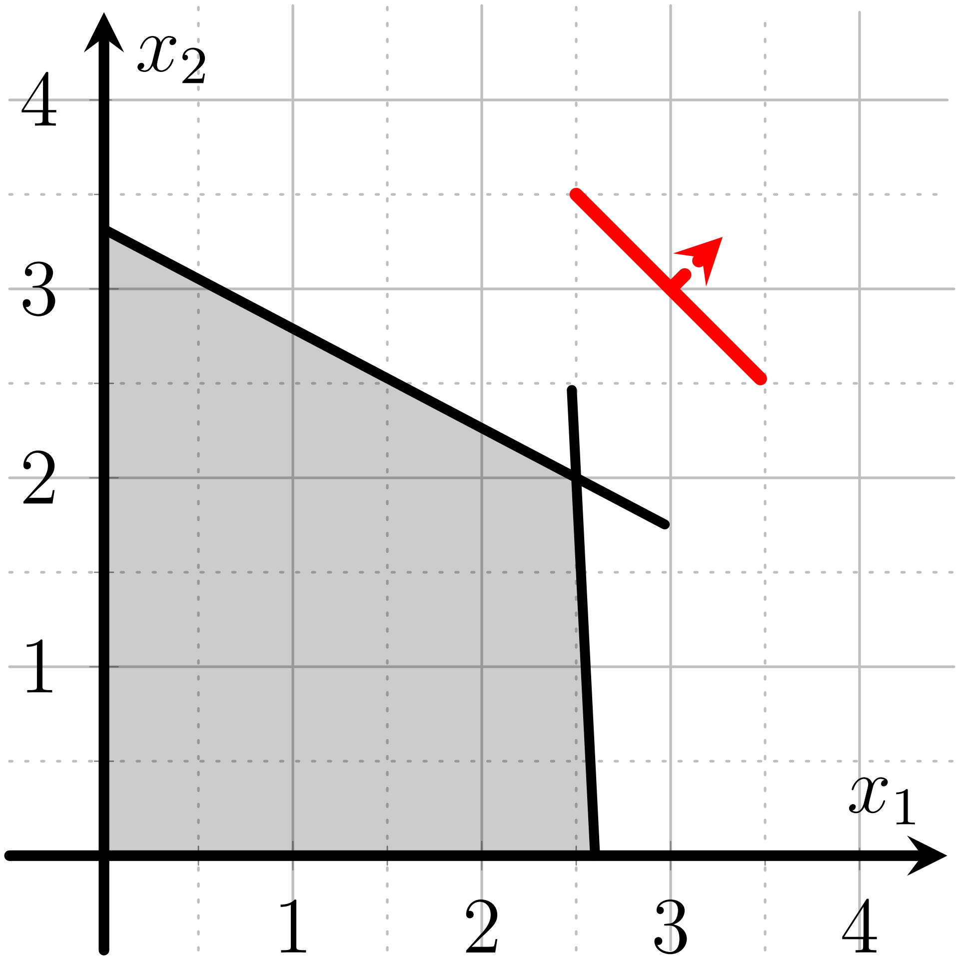
Consider the following *maximization problem*, given as an integer program (P)

$$\begin{array}{llrcl} (P) & \underset{\text{s.t.}}{\text{Max}} & x_1 + x_2 & & \\ & & 10x_1 + 19x_2 & \leq & 63 \\ & & 10x_1 + 0.5x_2 & \leq & 26 \\ & & x_1, x_2 & \in & \mathbb{N}_0 \end{array}$$

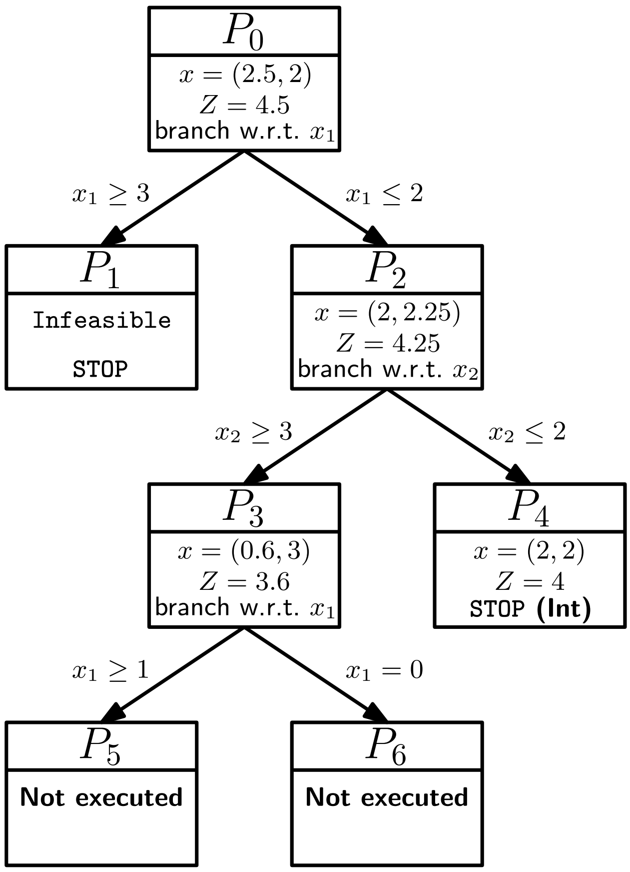
Solve the integer linear program (P) using the branch-and-bound method. Specify the optimal values for x_1 and x_2, as well as the optimal value of the objective function.

At each node, label the "greater-or-equal" branch with a lower subscript and the "less-than-or-equal" branch with a higher subscript. Process subproblems in first-in-first-out *FIFO* order.

Note: In the following coordinate system, the feasible set of LP relaxation of (P) and the objective function are shown. You can use it to solve the subproblems graphically.



Solution



A feasible integer solution is first found in P_4 with $Z = 4$. The program P_3 has $Z = 3.6$, so neither P_5 nor P_6 can lead to better objective function value, so the algorithm stops early.

Exercise D6. 2. *Cutting planes*

Problem

Given the following IP:

$$\begin{array}{rrcl} (P)\ \underset{\text{s.t.}}{\text{Max}} & 4x_1 + 2x_2 & & \\ & 2x_1 + 3x_2 & \leq & 3 \\ & -2x_1 - 2x_2 & \geq & -5 \\ & x_1, x_2 & \in & \mathbb{N}_0 \end{array}$$

a) Give the LP relaxation of (P) in canonical form.

b) Solve the LP relaxation using the simplex method and give the optimal solution and the optimal objective function value.

c) Give the inequality of a possible Gomory cut with respect to the determined optimal tableau.

Solution

a) The LP relaxation in canonical form is

$$\begin{array}{lllllcl} \underset{\text{s.t.}}{\text{Max}} & 4x_1 & +2x_2 & & & & \\ & 2x_1 & +3x_2 & +x_3 & & = & 3 \\ & 2x_1 & +2x_2 & & +x_4 & = & 5 \\ & & & x_1, x_2 & x_3, x_4 & \geq & 0. \end{array}$$

b) We can enter the LP directly into the simplex table and get

Base	Coefficients				Res.
	x_1	x_2	x_3	x_4	
z	-4	-2	0	0	0
x_3	2	3	1	0	3
x_4	2	2	0	1	5

$x_1 \leftrightarrow x_3$:

Base	Coefficients				Res.
	x_1	x_2	x_3	x_4	
z	0	4	2	0	6
x_1	1	1.5	0.5	0	1.5
x_4	0	-1	-1	1	2

The optimal solution of the LP relaxation is thus $(x_1, x_2) = (1.5, 0)$ with $Z = 6$.

c) There is only one base variable with a non-integer value. The defining equation is

$$x_1 + 1.5x_2 + 0.5x_3 = 1.5$$

If we split this into an integer part and a remainder and rearrange the equations accordingly, we get

$$x_1 + x_2 - 1 = -0.5x_2 - 0.5x_3 + 0.5$$

The Gomory cut is

$$x_1 + x_2 - 1 \leq 0 \quad \text{or} - 0.5x_2 - 0.5x_3 + 0.5 \leq 0$$

Exercise D6.3. *Cut the Planes*

Problem

The aircraft manufacturer *Airtruck* has developed two new models, the *Airtruck A381* and the *Airtruck A382*. Since production relies on limited resources, the company must decide how many units of each model to build in order to maximise profit.

This decision is modelled with the following integer programme:

$$\max z = x_1 + x_2 \tag{1}$$
$$\text{s.t. } x_1 + 2x_2 + s_1 = 5 \tag{2}$$
$$3x_1 + x_2 + s_2 = 6 \tag{3}$$
$$x_1 + s_3 = 2.4 \tag{4}$$
$$x_2 + s_4 = 3.8 \tag{5}$$
$$s_1, s_2, s_3, s_4 \geq 0 \tag{6}$$
$$x_1, x_2 \in \mathbb{N}_0 \tag{7}$$

After solving the LP relaxation with the simplex method, the optimal tableau is:

Base	Coefficients						Res.
	x_1	x_2	s_1	s_2	s_3	s_4	
z	0	0	2/5	1/5	0	0	16/5
x_1	1	0	$-1/5$	2/5	0	0	7/5
x_2	0	1	3/5	$-1/5$	0	0	9/5
s_3	0	0	1/5	$-2/5$	1	0	1
s_4	0	0	$-3/5$	1/5	0	1	2

a) Identify the basic variables that can be used to generate Gomory fractional cuts, and explain why they are eligible.

b) The company tests both Gomory cuts. Derive **all** valid representations of **each** Gomory cut from the tableau above.

c) Simplify the Gomory cut obtained from the x_1-row so that it can be directly incorporated into the integer programme (use the form without slack variables).

d) Formulate the updated integer programme including the first Gomory cut. You may reference earlier equations by their number (e.g., (1)) instead of rewriting them.

Solution

a) Only the basic variables x_1 and x_2 can be used to derive Gomory cuts because they have non-integer values in the optimal tableau.

b) **Cut from the x_1-row:** From the tableau we have

$$x_1 - \tfrac{1}{5}s_1 + \tfrac{2}{5}s_2 = \tfrac{7}{5}.$$

Splitting into integer and fractional parts:

$$x_1 - s_1 + \tfrac{4}{5}s_1 + \tfrac{2}{5}s_2 = 1 + \tfrac{2}{5}.$$

Rearranging gives

$$x_1 - s_1 - 1 = \tfrac{2}{5} - \tfrac{4}{5}s_1 - \tfrac{2}{5}s_2.$$

Thus, the Gomory cut can be written either as

$$x_1 - s_1 - 1 \leq 0 \quad \text{or} \quad \tfrac{2}{5} - \tfrac{4}{5}s_1 - \tfrac{2}{5}s_2 \leq 0.$$

Cut from the x_2-row: From the tableau we have

$$x_2 + \tfrac{3}{5}s_1 - \tfrac{1}{5}s_2 = \tfrac{9}{5}.$$

Splitting into integer and fractional parts:

$$x_2 - s_2 + \tfrac{3}{5}s_1 + \tfrac{4}{5}s_2 = 1 + \tfrac{4}{5}.$$

Rearranging gives

$$x_2 - s_2 - 1 = \tfrac{4}{5} - \tfrac{3}{5}s_1 + \tfrac{1}{5}s_2.$$

Thus, the Gomory cut can be expressed as

$$x_2 - s_2 - 1 \leq 0 \quad \text{or} \quad \tfrac{4}{5} - \tfrac{3}{5}s_1 + \tfrac{1}{5}s_2 \leq 0.$$

c) Substituting back into the original constraints, with $s_1 = 5 - x_1 - 2x_2$, we obtain

$$x_1 - (5 - x_1 - 2x_2) - 1 \leq 0 \quad \Rightarrow \quad 2x_1 + 2x_2 \leq 6 \quad \Leftrightarrow \quad x_1 + x_2 \leq 3.$$

d) The updated LP with the cut from x_1 is:

$$\begin{array}{ll} & (1) \\ \text{s.t.} & (2), (3), (4), (5), (7) \\ & \mathbf{2x_1 + 2x_2 + s_5 = 6} \\ \text{or} & \mathbf{x_1 + x_2 + s_5 = 3} \\ & s_1, s_2, s_3, s_4, \mathbf{s_5} \geq 0 \\ \text{or} & (6) \text{ and } \mathbf{s_5 \geq 0} \end{array}$$